Humanities & Social Sciences Communications

ARTICLE

https://doi.org/10.1057/s41599-025-05641-x OPEN

Artificial intelligence, social influence, and AI anxiety: analyzing the intentions of science doctoral students to use ChatGPT with PLS-SEM

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The widespread use of AI-based applications, such as ChatGPT, in higher education has raised questions about how doctoral students adopt these technologies, particularly in science departments. However, traditional models, such as the Technology Acceptance Model (TAM), may fall short in fully capturing users' behavioral intentions toward emerging technologies. This study aims to investigate PhD students' behavioral intention to use ChatGPT for learning purposes by proposing an extended version of the TAM. The proposed model incorporates five additional factors: Social Influence (SI), Perceived Enjoyment (PEN), AI Self-Efficacy (AI-SE), AI's Sociotechnical Blindness (AI-STB), and Perceived Ethics (AI-PET). Data were collected from 361 PhD students across 35 universities and analyzed using Partial Least Squares Structural Equation Modeling (PLS-SEM). The results reveal that all proposed factors significantly influence students' behavioral intention to use ChatGPT. This study contributes to the literature by introducing new external variables into the TAM framework and providing empirical insights into doctoral students' acceptance of AI tools in science education, thereby offering a novel perspective for future research and educational practice.

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Introduction

Artificial intelligence (AI) is defined as a broad field that aims to develop systems that mimic human intelligence (Collins et al. 2021). This technology, which offers human-like or more advanced capabilities, has been developed in response to the question "Can machines think?" posed by Alan Turing in the 1950s (Tlili et al. 2023; Taani and Alabidi, 2024). This fundamental question played an important role in the evolution of AI (Adamopoulou and Moussiades, 2020), leading to the development of the first chatbots, such as "ELIZA (Weizenbaum, 1966)", "PARRY (Colby et al. 1971)", "ALICE (Wallace, 2009)", and "SmarterChild (Molnar and Szuts, 2018)"(Zhou et al. 2024). Over time, the functionality of chatbots has improved significantly, evolving into more complex and intelligent systems that enhance the user experience. Chatbots are now characterized as advanced AI systems that simulate human-like speech through text or voice interactions (Camilleri, 2024). These systems generate natural language responses based on user input using algorithms trained on large data sets (Menon and Shilpa, 2023). In higher education institutions (HEIs I), AI and AI tools, especially advanced language models like ChatGPT, play a critical role in teaching and learning processes and are widely discussed (Lo, 2023; Tupper et al. 2024; Lo et al. 2024; Wang et al. 2024c). Artificial Intelligence in Education (AIED) is "transforming teaching and learning processes", "providing innovative learning experiences," and "enhancing the effectiveness of educational management" (Duong et al. 2023; Gidiotis and Hrastinski, 2024; Saihi et al. 2024). AIED supports this transformation with functions such as monitoring student performance, identifying individual deficiencies, and providing customized feedback (Montenegro-Rueda et al. 2023; Gidiotis and Hrastinski, 2024; Saihi et al. 2024). Chatbots enhance students' self-regulation skills (Montenegro-Rueda et al. 2023), provide insights into student behaviors to improve learning outcomes (Kuhail et al. 2023; Alhwaiti, 2023; Liu and Ma, 2024), and play a significant role in the presentation of course content (Taani and Alabidi, 2024).

These technologies support cognitive development (Bhullar et al. 2024), reduce learning anxiety (Chiu et al. 2023), facilitate second language learning (Karataş et al. 2024), and enhance programming skills (Yilmaz and Karaoglan Yilmaz, 2023). In addition, chatbots contribute to developing critical thinking and writing skills (Tupper et al. 2024), and career planning (Bhutoria, 2022). However, the use of AI chatbots in education also raises ethical, moral, and security issues (Stahl and Eke, 2024). Trustworthy AI should be based on principles such as human oversight, transparency, non-discrimination, and accountability (Díaz-Rodríguez et al. 2023). AIED applications face challenges such as ethical issues, inadequate evaluation, and user attitudes (Okonkwo and Ade-Ibijola, 2021), and non-transparent decision-making processes (Bathaee, 2018). These issues may limit the credibility and use of AI. In addition, the overuse of AI raises concerns that it may negatively affect users' independent thinking skills (Qin et al. 2020). Despite these concerns, AI technologies can potentially revolutionize the academic world through innovative applications in education (Rawas, 2024).

ChatGPT, a large language model (LLM) based chatbot developed by OpenAI, has brought a critical perspective to teaching and learning practices in higher education (Bhullar et al. 2024) and has gained an important place (Habibi et al. 2023). The ChatGPT model, released in May 2024, stands out for its features that enable more natural and fluent communication in human-computer interaction (OpenAI, 2024). ChatGPT is a robust and reliable Natural Language Processing (NLP) model that can understand and generate natural language with high accuracy in many domains, such as text generation, language comprehension, and development of interactive applications (Gill and Kaur, 2023). The model's competencies in contextual understanding, language generation, task adaptability, and multilingualism provide significant advances in the field of GenAI (Ray, 2023).

The impact of tools such as AI and ChatGPT on the learning processes of HEIs students is increasingly being explored (Lo, 2023; Strzelecki, 2023; Tayan et al. 2024; Bhullar et al. 2024; Lo et al. 2024). GenAI tools that offer quality human-like interactions, especially large language models such as ChatGPT, are found to be more useful and acceptable to learners. As Lin and Yu (2025a) state, "the perceived trustworthiness and human-like features of AI-generated pedagogical agents have a significant impact on students' perceived usefulness and intention to use" (p. 1620). Several studies have examined university students' behavioral intentions regarding using artificial intelligence tools in learning and developed policies and strategies using different models (Dahri et al. 2024; Al Shloul et al. 2024; Grassini et al. 2024). Conceptual models are used to explain individuals' acceptance of new technologies. These models are generally based on individuals' attitudes, intentions, and behaviors (Davis, 1989). The Technology Acceptance Model (TAM) is a widely utilized theoretical framework in the academic literature to elucidate the factors influencing individuals' technology acceptance and the processes through which they accept new technologies (Davis and Venkatesh, 1996; Almogren et al. 2024; Saihi et al. 2024). TAM is considered an approach that provides researchers with important information to predict users' behavioral intentions, especially regarding the use of technology decisions (Marangunić and Granić, 2015). However, due to the multidimensional nature of behavioral intentions toward AI technologies (Bergdahl et al. 2023; Zhang et al. 2024b), it is suggested in the literature that TAM does not form an ideal structure for determining user acceptance (Schepman and Rodway, 2020). In addition, some psychological factors may play a more dominant role in accepting AI than in accepting other technologies (Bergdahl et al. 2023). It can be concluded that some new external factors still need to be considered to explain the users' behavioral intentions. Furthermore, there are very few studies on whether and how students accept ChatGPT for learning purposes. Specifically, there is a significant gap in the literature regarding how PhD students adopt these technologies. While AI-based chatbots have shown great potential in writing, little is known about whether and how PhD students accept the use of ChatGPT (Zou and Huang, 2023).

In light of the consideration given above, the motivation of this study comes from extending the TAM by jointly including new external factors such as Social Influence (SI), Perceived Enjoyment (PEN), AI Self-Efficacy (AI-SE), AI's Sociotechnical Blindness (AI-STB), and AI Perceived Ethics (AI-PET) in addition to the factors of Perceived Usefulness (PU), Perceived Ease of Use (PEOU), and Behavioral Intention to Use (BI) already present in the TAM. Additionally, this study investigates the behavioral intentions of PhD students in natural sciences toward ChatGPT using the proposed model. In the application part, 361 PhD students from various scientific fields in natural sciences at 35 universities participated in the study. The data were analyzed using PLS-SEM.

This study offers a novel contribution by extending the TAM with AI-specific sociotechnical and ethical constructs to explain doctoral students' behavioral intention to use ChatGPT for teaching and learning in science education.

Study objectives

This study aims to achieve the following objectives:

1. To analyze the impact of these newly proposed external factors on the behavioral intention of PhD students in the natural sciences toward adopting ChatGPT.

2. To develop and test an extended version of the TAM that includes Social Influence (SI), Perceived Enjoyment (PEN), AI Self-Efficacy (AI-SE), AI's Sociotechnical Blindness (AI-STB), and AI Perceived Ethics (AI-PET) to explain PhD students' behavioral intention to use ChatGPT for learning purposes.

ChatGPT in education

ChatGPT, with its human-like text generation and natural responsiveness (Jo and Park, 2024) transforms traditional teaching methods and provides students with personalized learning experiences (Tlili et al. 2023; Grassini et al. 2024). The main motivations for using ChatGPT among university students include quick response, ease of use, and accessibility (Klimova and de Campos, 2024). The use of ChatGPT in different disciplines varies depending on students' familiarity with technology and perceived benefits (Romero-Rodríguez et al. 2023). ChatGPT and AI tools are found to be used more intensively by students in engineering, mathematics, and science fields due to their proximity to technology (von Garrel and Mayer, 2023). ChatGPT is gaining popularity in computer science and engineering, especially for skills such as writing, modifying, and debugging code (Geng et al. 2023; Kosar et al. 2024; Lo et al. 2024). In science, ChatGPT has a significant impact on knowledge analysis and understanding of complex concepts. ChatGPT responses to chemistry questions provide students with a better understanding of the topics (Fergus et al. 2023; Guo and Lee, 2023), while civil engineering students report that ChatGPT responses are more comprehensive and informative (Uddin et al. 2024). In modern statistics education, ChatGPT has been observed to help students with technical subjects such as *R* programming (Ellis and Slade, 2023; Xing, 2024). However, in subjects such as geometry, it has been reported that ChatGPT does not have a deep enough understanding, it is difficult to correct misconceptions, and its accuracy can vary depending on the complexity of the prompts given (Wardat et al. 2023).

The generative and iterative processes associated with artificial intelligence generated content (AIGC) require learners not only to evaluate the utility of these technologies but also to align their cognitive strategies with the ever-evolving capabilities of AIGC (Wang et al. 2025). Wang et al. (2025) stated that students need to interact with the outputs of generative AI tools such as ChatGPT, critically evaluate and reprocess these outputs; this process requires them to restructure their cognitive strategies in line with the technology.

Evaluations of the positive and negative effects of ChatGPT in higher education have been discussed with increasing interest in the academic world in recent years (Lo, 2023; Strzelecki, 2023; Tayan et al. 2024; Lo et al. 2024). Although ChatGPT has many potential benefits, such as enhancing the learning experience, speeding up administrative tasks (Tayan et al. 2024), increasing student engagement, and supporting adaptive learning processes (Wang et al. 2024b), research on the factors having an impact on using this technology and the causal relationships between these factors is limited (Baig and Yadegaridehkordi, 2024). Students' acceptance and use of ChatGPT are affected by factors called perceived usefulness, perceived convenience, and familiarity with the technology. Furthermore, considering gender as an important variable in technology use (Gefen and Straub, 1997; Venkatesh and Morris, 2000; Arthur et al. 2025), research suggests that male students are more likely to use AI chatbots (Bastiansen et al. 2022; Stöhr et al. 2024).

In addition to its benefits, ChatGPT raises important ethical concerns such as social justice, individual autonomy, cultural identity, and environmental issues (Stahl and Eke, 2024). One of these concerns is that biased or outdated information in the datasets used for training ChatGPT may reproduce social and cultural biases, which may reduce the credibility of academic research (Farrokhnia et al. 2024). In the academic community, the ethical dimensions of ChatGPT are increasingly being debated, with issues such as the accuracy of the content, the impact on academic integrity, and the ethical use of AIED being among the main concerns (Cooper, 2023). Also, the high learning value provided by generative AI may increase the tendency of university students to move away from processes that require cognitive effort, and this may lead to a decrease in students' willingness to use independent thinking skills over time (Tang et al. 2025).

Especially in thesis writing, exams, and assignments, the production of unverified content without citing sources or using original thinking (Acosta-Enriquez et al. 2024a) increases the risk of plagiarism and causes serious problems in terms of academic integrity (Lo, 2023; Chaudhry et al. 2023). The ease of use that ChatGPT offers in tasks ranging from text generation to coding (Stojanov, 2023), may lead students to use information directly without engaging their analytical thinking and decision-making skills (Abbas et al. 2024). This leads students to neglect the skills of content validation, critical evaluation, and ethical behavior, resulting in a superficial approach to knowledge and a lack of academic integrity. The risk of producing false or biased information misleads students and undermines trust in educational processes (Rahimi and Talebi Bezmin Abadi, 2023; Qadir, 2023).

ChatGPT misuse threatens academic integrity in higher education (Memarian and Doleck, 2023; Geerling et al. 2023; Espartinez, 2024), raising issues of plagiarism, academic dishonesty (Farhi et al. 2023) accountability, and confidentiality. However, "over-reliance on ChatGPT" and its potential consequences should be considered multidimensionally. Although ChatGPT increases student engagement in the learning process, its unconscious or excessive use has been reported to lead to cognitive overload (Woo et al. 2024). Unconscious use of this technology can negatively impact students' critical thinking and problem-solving skills (Kasneci et al. 2023); students may miss opportunities to develop their knowledge and skills by focusing on academic success with pre-determined answers (Tlili et al. 2023). This situation may increase students' reliance on technology, which in turn can reduce their capacity for independent thinking and problem-solving over time (Zhang et al. 2024a; Zhai et al. 2024).

Theoretical framework and research hypotheses

Technology acceptance model. The integration of new technologies, especially AI technologies, and tools, in the field of education and training (Elbanna and Armstrong, 2024) is receiving increasing attention. The education system has a large audience of students who can use AI technologies in the process of knowledge transfer and acquisition. Consequently, an understanding of the manner in which users accept AI technologies and an analysis of the adoption processes are of critical importance for the effective use of these technologies in an educational context. The TAM is a powerful model (Sagnier et al. 2020; Lim and Zhang, 2022) that has been widely used to study the acceptance of AI technologies and provides a comprehensive framework for understanding users' cognitive and behavioral responses to technologies (Granić and Marangunić, 2019).

TAM, developed by Davis (1989), is considered one of the most effective models for examining what triggers users' adoption of technology (Koenig, 2024). TAM was developed to understand the relationship between the characteristics of a technology and users' adoption of that technology and emphasizes the mediating

role of two key factors in this process, e.g., PEOU and PU (Marangunić and Granić, 2015).. These factors are important determinants shaping users' cognitive and behavioral responses to technology.

TAM reflects users' individual preferences for technology use; however, in complex technologies such as AI, apart from "individual preferences, social norms, other' decisions" (Schepman and Rodway, 2020, p. 11), and various psychological factors (Bergdahl et al. 2023) play an important role. Sohn and Kwon (2020) argue that more than the TAM model may be required to explain innovative technologies, especially AI-based applications. Although TAM has been criticized for its repetitive use, it remains a valid and widely applied framework, especially when extended with context-specific variables. In this study, TAM was enriched with AI-related constructs to better reflect the dynamics of generative AI adoption, offering a simplified yet robust model suitable for the research context. While UTAUT and UTAUT2 offer comprehensive frameworks for understanding technology acceptance, their structural complexity and reliance on multiple moderating variables may limit their applicability in studies aiming for model parsimony and context-specific insights. Therefore, the extended TAM framework has been used in many studies to predict student acceptance of AI technologies and ChatGPT (Al-Adwan et al. 2023; Baig and Yadegaridehkordi, 2024; Saihi et al. 2024). Important studies examining university students' use of AI tools and ChatGPT within the TAM framework are given in Table 1 below.

Studies examining HIE students' intentions and behaviors toward the use of ChatGPT have not only been limited to TAM but also studies using other technology acceptance models, such as UTAUT2 with modifications (Foroughi et al. 2023; Habibi et al. 2023; Strzelecki, 2023; Bouteraa et al. 2024; Arthur et al. 2025). As shown in Table 1, the theoretical framework of the classical TAM model has been extended by adding various exogenous factors according to the characteristics of the studies. Al-Abdullatif (2023) combined the relationships between the constructs of TAM and the value-based model (VAM) to examine how students perceive and accept chatbots in their learning process. The results showed that PU, PEOU, attitude, perceived enjoyment, and perceived value were the determinants of students' acceptance of chatbots. However, the perceived risk variable did not significantly affect students' attitudes or acceptance of using chatbots for learning. Lai et al. (2023), in a study conducted with 50 undergraduate students from eight universities in Hong Kong, examined the role of intrinsic motivation in the intention to use ChatGPT and added only the intrinsic motivation variable to the classical TAM model. The research results show that PU strongly predicts behavioral intention, but there is no significant relationship between PEOU and behavioral intention (BI). The results also show that PU and PEOU do not mediate considerably in the relationship between intrinsic motivation and BI.

Zou and Huang (2023) examined PhD students' acceptance of ChatGPT in writing and the factors influencing it within the framework of the TAM; it was found that there is a high intention among PhD students in China to use ChatGPT in writing. Bilquise et al. (2024) combined traditional TAM and UTAUT models with AI-driven self-service technology models and sRAM service robot acceptance models to examine UAE university students' acceptance of ChatGPT for academic support within the Self Determination Theory (SDT) framework. Similar to previous research, this study shows that PU, PEU, trust, and social efficacy are important factors in acceptance. This reemphasizes the importance of the combined influence of individual and social factors in AI-based technologies. Similarly, Saihi et al. (2024) emphasized that PU and perceived efficacy factors have strong positive effects on the intention to adopt AI chatbots, but concerns about "trust and privacy," "interaction," and "information quality" are evident. Sallam et al. (2024) in their study conducted in higher education institutions in the United Arab Emirates found that students' positive attitudes towards the use of ChatGPT were associated with gender (male), Arab origin, low GPA and PU, low perceived risk of use, and high cognitive structures and PEOU factors. These findings suggest that individual demographic characteristics and perceptual factors may determine acceptance of AI-based technologies. Almogren et al. (2024) in a study conducted with university students in Saudi Arabia, added seven exogenous factors to the TAM model and found that PEOU and PU were important determinants in explaining attitudes toward ChatGPT. The study showed that subjective norms did not significantly affect attitudes toward the educational use of ChatGPT. The results emphasize the importance of individual perceptions in technology acceptance by showing that individual perceptions, ease of use, and feedback quality are more effective than social pressure.

Liu and Ma (2024), in their study of EFL (English as a foreign language) students, found that perceived enjoyment (PEN) and attitude were strong determinants of ChatGPT adoption; however, PEOU did not directly affect attitude. Nevertheless, PEOU significantly affected PU, which mediated users' development of positive attitudes and behaviors toward the technology. These findings suggest that improving the user experience can promote the adoption of ChatGPT in educational settings. In a study conducted with 108 students from public and private universities in Vietnam, it was found that PEOU influenced intention to accept ChatGPT, while PU had no direct effect but an indirect effect through personalization and interactivity (Maheshwari, 2024). This finding suggests that personalizing the user experience and providing interactive elements can increase the acceptance of AI-based tools like ChatGPT. Boubker (2024) found that in the acceptance of ChatGPT by students of higher education institutions in Morocco, output quality impacts PU, usage, and student satisfaction, and social influence also plays an important role in this process. Chang et al. (2024) investigated the differences between students with weak and strong digital AI skills, using a latent class model to assess the digital AI skills of college students. The results showed that attitude and perceived behavioral control played important roles in ChatGPT acceptance; furthermore, subjective norms were also effective in ChatGPT acceptance among students with strong AI digital abilities. These findings indicate that digital literacy is a critical factor in the technology adoption process. Güner et al. (2024), in their study of higher education students in Turkey, found that perceived enjoyment had a positive effect on PU, PEOU, and attitude toward ChatGPT use; however, personal innovativeness had a limited effect on attitude.

In this study, as similar to the literature (see Table 1), an extended TAM model was used to predict the acceptance of ChatGPT by PhD students in the field of science. In the extended model, in addition to the PU and PEOU variables of the classical TAM,

External predictors: The external predictors used to predict perceived usefulness and perceived ease of use include AI's Sociotechnical Blindness (AI-STB), AI Perceived Ethics (AI-PET), and AI's Perceived Self-Efficacy (AI-SE).

Factors from other theories: To increase the predictive validity of TAM, Social Influence (SI) and Perceived Enjoyment (PEN) factors were included, and the model was structured according to these two main categories.

SI and PEN encourage users to use technology more motivated, while AI-SE increases users' trust in technology. AI-STB indicates that users can ignore the possible negative effects of AI, while

Table 1 Review of Literature.

References	*Non-TAM variables*	*Respondents*	*Aim*
Al-Abdullatif (2023)	Perceived enjoyment, Perceived value, Perceived risk, and Chatbot Acceptance.	Undergraduate and graduate students in Saudi Arabia	By examining the interaction of TAM and VAM factors, it aims to predict the acceptance of chatbots like ChatGPT and Bard among students.
Algerafi et al. (2023)	Subjective norm, Image, Results demonstrability, Job relevance, Output quality, Anxiety, Self-efficacy, Perceived enjoyment, and Perceived playfulness.	HEIs students in China	The emphasis has been on assessing students' intent to use AI-based robots in the classroom using the TAM 3 model.
Bilquise et al. (2024)	Social Influence (SI), Perceived Trust (PT), Perceived Autonomy (PA), and anthropomorphism (AN).	HEIs students in the UAE	Based on a conceptual model derived from TAM, UTAUT, Service Robot Acceptance (sRAM), and Self Determination Theory (SDT), it examines the functional, socio-emotional, and relational factors affecting students' acceptance of the academic advising chatbot.
Lai et al. (2023)	Intrinsic motivation.	Undergraduate students in Hong Kong	It attempts to extend the TAM to facilitate active learning and investigate the role of intrinsic motivation in adopting ChatGPT.
Liu and Ma (2024)	Classic TAM	EFL learners in China	It empirically examines how ChatGPT is perceived and used by EFL students outside the classroom.
Maheshwari (2024)	Perceived trust, Interactivity, Personalization, and Perceived Intelligence.	Undergraduate and graduate students in Vietnam	It examined the factors affecting students' intention to adopt and use ChatGPT in academic studies.
Zou and Huang (2023)	Experience.	Doctoral Students in China	Within the framework of TAM, the acceptance of ChatGPT among doctoral students for writing purposes and the factors influencing this acceptance have been examined.
Rahman et al. (2023)	Perceived informativeness, Perceived enjoyment, and Trust.	University students in Bangladesh	It aims to understand the factors that trigger students to use ChatGPT for learning purposes.
Shaengchart (2024)	Privacy, Security, and Facilitating Conditions.	Higher education students in Bangkok, Thailand	A series of hypotheses have been tested to understand the key factors affecting the adoption of ChatGPT.
Tiwari et al. (2024)	Perceived credibility, Perceived Social Presence, and Hedonic motivation.	HEIs students in Oman	Based on the TAM, the aim is to identify the factors determining students' attitudes toward using ChatGPT for educational and learning purposes.
Abdaljaleel et al. (2024)	Perceived risk of use, General perceived risk, Anxiety, and Technology/Social influence.	HEIs students in Arab Countries	It aimed to examine the factors affecting university students' attitudes towards and usage of ChatGPT in Arab countries.
Abdalla (2024)	Awareness, Social Influence, and Perceived Enjoyment.	College Student in Palestine	Factors affecting the intention to use ChatGPT among university students have been investigated through an extended TAM model, with personalization considered as a moderator.
Almogren et al. (2024)	Trialability, Perceived compatibility, Relative advantage, Trust in ChatGPT, Feedback quality, Perceived assessment Quality, and Subject norms.	University students from different disciplines in Saudi Arabia	It has investigated the factors that shape users' attitudes, intentions, and behaviors toward adopting ChatGPT for smart education systems.
Albayati (2024)	Privacy, Social Influence, Security, and Trust.	Undergraduate students in Korea	It evaluates the factors affecting users' acceptance of ChatGPT as a daily reference tool and the different user awareness levels.
Almulla (2024)	Interaction learning (INL), Interaction quality (IQ), Collaborative learning (CL), Information Quality (INQ), learning motivation (LM), and learning satisfaction (LS).	HEIs students in Saudi Arabia	The factors affecting university students' adoption of ChatGPT for research purposes and the impact of this usage on learning satisfaction have been examined.
Boubker (2024)	Output quality, social influence, Student satisfaction, and Individual impact.	HEIs students in Morocco	The role of ChatGPT in enhancing students' learning outcomes has been investigated.
Dahri et al. (2024)	Personal competence, social influence, Perceived AI trust, Perceived AI usefulness, Perceived AI enjoyment, Perceived AI intelligence, and Metacognitive self-regulation learning.	Preservice teachers in Malaysia	The adoption of ChatGPT by teacher candidates for self-regulated learning has been examined.
Chang et al. (2024)	Subjective norm.	Undergraduate, master, and PhD students in China	The acceptance of ChatGPT has been tested among different student groups.

Table 1 (continued)

References	*Non-TAM variables*	*Respondents*	*Aim*
Güner et al. (2024)	Perceived enjoyment, Personal innovativeness, Internet privacy, and social influence.	Undergraduate students in Türkiye	University students' attitudes and perspectives towards using ChatGPT in their courses have been examined.
Ma and Lei (2024)	Artificial Intelligence Literacy (AIL), Subjective Norms (SN), and Output Quality (OQ).	Pre-service teacher education students in China	The effects on the willingness of teacher education students to adopt AI technologies have been examined.
Saif et al. (2024)	Anxiety.	Last year (MBA) students in the discipline of Management Sciences	The study aimed to examine the use of ChatGPT with ubiquitous learning (UL) within the TAM framework, focusing on the relationship between students' stress and anxiety in academic tasks and technology integration.
Saihi et al. (2024)	Perception Factor, Trust and Privacy Factor, Interaction and System Factor, Social and Contextual Factor, and Outcome Variables.	HEIs students and educators in UAE	The complex dynamics shaping the perceptions of educators and students towards AI chatbots (ChatGPT) have been examined.
Sallam et al. (2024)	Behavior/cognitive factors, Perceived risk, Anxiety, and Technology/social influence.	HEIs students in UAE	It aimed to evaluate the factors affecting university students' attitudes and usage of ChatGPT.

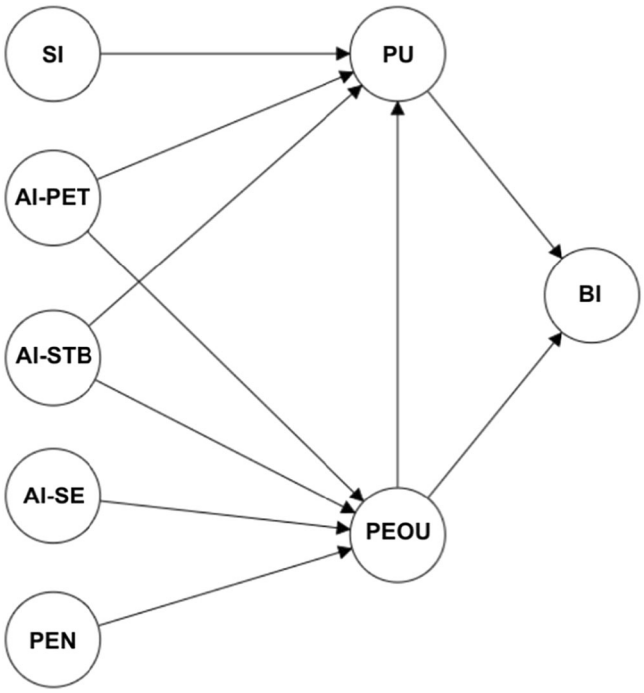

Fig. 1 The conceptual research model.

AI-PET stands out as a critical factor for users to develop trust in AI technology. The extended model proposed in this study is illustrated in Fig. 1.

Hypotheses development

Social Influence (SI). Social Influence (SI), which plays an important role in the early acceptance of new technologies (Venkatesh et al. 2003), is one of the determining factors in using AI technologies for learning purposes. The Theory of Reasoned Action (TRA), one of the main theoretical underpinnings of the original development of TAM, addresses social influence through subjective norm (Venkatesh and Davis, 2000) Subjective norm is defined by Ajzen and Fishbein (1980) as "a person's perception that most people important to him or her think that he or she should or should not perform a particular behavior" (Venkatesh and Davis, 2000, p. 187). Venkatesh and Davis (2000) found that SI significantly affects perceived usefulness through internalization when people incorporate social influences into their perceptions of usefulness and use a system to improve their job performance. In this study, SI was defined as graduate students' beliefs about the encouragement and support they receive from those around them (faculty members, peers, etc.) regarding the use of ChatGPT (Ayaz and Yanartaş, 2020; Venkatesh, 2022). Various studies (Algerafi et al. 2023; Zhang et al. 2023; Boubker, 2024; Dahri et al. 2024; Albayati, 2024; Abdalla, 2024) emphasized that SI has an effect on PU. Consequently, it is assumed that graduate students' beliefs about the usefulness of ChatGPT are positively influenced by their social environment.

H1: SI positively influences the PU of ChatGPT.

Perceived Enjoyment (PEN). PEN, which influences the acceptance of new technologies as an intrinsic motivator (Davis et al. 1992), refers to users' beliefs that they can enjoy using a particular technology (Venkatesh, 2000). The positive effect of perceived enjoyment of learning technology on PEOU and BI has been confirmed in studies in the literature (Zhou et al. 2022; Algerafi et al. 2023; Zhang et al. 2023; Güner et al. 2024; Dehghani and Mashhadi, 2024; Abdalla, 2024). In this framework, it is expected that there is a strong relationship between PEN and PEOU (Davis et al. 1992) for PhD students using ChatGPT as a learning tool.

H2: PEN positively influences the PEOU of ChatGPT.

AI Self-Efficacy (AI-SE). Self-efficacy, which expresses an individual's belief in their ability to succeed in a specific task (Bandura, 1982), in the AIED perspective, refers to the general beliefs about an individual's ability to interact with AI, successfully learn AI concepts and tools, and apply them (Wang and Chuang, 2024; Kim and Kim, 2024). AI-SE is considered a strong predictor of learning and success (Ng, 2023). It has been reported in many studies that individuals with high self-efficacy perceive the use of new technologies as easier (Granić, 2023; Al Darayseh, 2023; Algerafi et al. 2023; Al-Adwan et al. 2023; Zhang et al. 2023; Güner et al. 2024; Pan et al. 2024). It has been assumed that PhD students have a high AI-SE due to their educational level; therefore, their perceptions of the ease of use of ChatGPT would be higher.

H3: AI-SE positively influences the PEOU of ChatGPT.

Perceived AI Ethics (AI-PET). Ethics is a moral principle determining the rightness or wrongness of specific actions (Raghunathan and Saftner, 1995). Ethics in the context of AIED are beliefs regarding the moral and ethical implications of using AI-based learning tools for academic purposes (Acosta-Enriquez et al. 2024b). AI-PET significantly influences student's behavioral intentions to accept and integrate ChatGPT into learning environments (Khechine et al. 2020; Farhi et al. 2023; Salifu et al. 2024), while severe ethical concerns negatively impact these behavioral intentions (Shao et al. 2024). Therefore, it is anticipated that PhD students who find the use of AI tools for learning purposes ethical will find the use of ChatGPT easy and beneficial.

H4: AI-PET will be positively associated with the PEOU of ChatGPT.

H5: AI-PET ethics will be positively associated with the PU of ChatGPT.

AI's Sociotechnical Blindness (AI-STB). It has been emphasized in various studies that the stress and anxiety caused by students' use of educational technologies in higher education has become an increasingly important problem (Lin and Yu, 2025b). "AI anxiety – AI-ANX" in its simplest definition, expresses individuals' fears and concerns about uncontrolled AI (Johnson and Verdicchio, 2017a, b). AI-ANX is seen as a significant precursor belief (Wang and Wang, 2022) influencing attitudes and external factors that determine behavioral intentions to accept AI tools (Chan, 2023; Wang et al. 2024b). Unlike other technological concerns, AI-ANX triggers, such as confusion regarding autonomy and sociotechnical (Johnson and Verdicchio, 2017a, b, 2024), are being discussed more extensively. Wang and Wang (2022) defined AI-ANX dimensions as learning, job replacement, sociotechnical blindness, and AI configuration. Sociotechnical blindness refers to an excessive focus on AI programs that obscures people's fundamental role at every stage of designing and implementing AI systems (Johnson and Verdicchio, 2017a). In the study, it is assumed that PhD students' beliefs about the misuse, malfunction, or potential consciousness and surpassing human control of ChatGPT will have a negative impact on their PEOU and PU.

H6: AI-STB negatively influences the PEOU of ChatGPT.

H7: AI-STB negatively influences the PU of ChatGPT.

Perceived Ease of Use (PEOU). PEOU refers to how easy it is perceived to use and learn a technology (Davis, 1989). The perception of ease of use enables individuals to find technology useful and increases their intention to use it (Venkatesh et al. 2003). The impact of PEOU on PU and BI regarding AI acceptance and adoption has been demonstrated in numerous studies (Alyoussef, 2023; Algerafi et al. 2023; Zhang et al. 2023; Pang et al. 2024; Shao et al. 2024; Almogren et al. 2024; Almulla, 2024; Abdalla, 2024). In this context, it is expected that doctoral students who perceive ChatGPT as easy to use will have a high perceived usefulness and stronger behavioral intentions.

H8: PEOU positively influences the PU of ChatGPT.

H9: PEOU positively influences BI to use ChatGPT.

Perceived Usefulness (PU). The PU reflects individuals' beliefs that using a particular system will enhance their job performance (Davis et al. 1989; Davis, 1989). PU significantly influences behavioral intentions as a strong determinant in the technology adoption process (Venkatesh et al. 2012; Alyoussef, 2023; Algerafi et al. 2023; Zou and Huang, 2023; Zhang et al. 2023; Güner et al. 2024; Pan et al. 2024; Abdalla, 2024). It is expected that PhD students' beliefs in the benefits of ChatGPT for learning will significantly influence their intention to use this technology.

H10: PU positively influences BI to use ChatGPT.

When the research hypotheses presented above are considered in their entirety, it is anticipated that the direct, indirect, and total effects between the five exogenous variables (SI, PEN, AI-SE, AI-STB, and AI-PET) incorporated into the model and the two endogenous variables (PU and PEOU) representing mediation will provide the most comprehensive explanation of the behavioral intentions of PhD students about the use of ChatGPT. The study's conceptual framework, shown in Fig. 1, was created based on the research hypotheses defined above.

Research method

Research design and sample. This study adopted a three-stage research design to minimize sampling bias and ensure the sample's representativeness. *In the first phase*, the target population consisted of doctoral students in the natural sciences (mathematics, physics, chemistry, biology, and statistics) at 35 universities in Türkiye. These universities, scoring between 20 and 60 points in the national TUBITAK university ranking (TÜBİTAK, 2023), were selected because they are known for high-quality education and are typically preferred by students who are more successful in university entrance exams and more interested in technology and artificial intelligence. *In the second phase*, using a Python tool specifically developed for the research, information (title, field, email) about academic staff working in the field of natural sciences at the corresponding 35 universities was collected from the YÖKAKADEMIC (YÖK, 2024a) database (a central system containing information and resumes of academic staff). These academic staff members were informed via email and requested to provide us with the email addresses of their doctoral students. Then, the email addresses of doctoral students sent by their advisors were listed randomly in a document with a line number. *In the third phase*, a Python tool was developed to generate a uniform random number between 1 and 651. Then, units with line numbers matching the generated random number via the Python tool were selected as sampling units, i.e., samples were chosen randomly by following the random sampling rules (De Muth, 2014, pp. 43–48).

As a result, the online survey was completed by 402 PhD students, who were selected randomly, between May and June 2024. However, 27 participants reported not using any AI tools, and 14 did not complete the survey questions; therefore, the final sample consisted of 361 PhD students. This sample size satisfies the requirement for detecting a minimum path coefficient of 0.15 with a 5% error probability, as suggested by Kock and Hadaya (2018) using the inverse square root method, which requires a minimum sample size of 275 (Sarstedt et al. 2021, p. 29) During the data collection process, no incentives were offered to participants; participation was entirely voluntary and conducted in accordance with ethical guidelines. The demographic characteristics of the participants are presented in Table 2.

Data collection

Research instruments. The study data were collected through a structured questionnaire consisting of two parts. The first part of the questionnaire included questions about the demographic characteristics of the participants, while the second part included scale items to assess doctoral students' behavioral intentions to use ChatGPT technology for educational purposes. These items were adapted from previously validated studies within the TAM framework and restructured to fit the context of the study. The second part of the questionnaire includes the following constructs: SI (4 items, Venkatesh et al. 2003), PEN (4 items, Venkatesh and Bala, 2008), AI-SE (4 items, Şahin et al. 2022; Salloum et al. 2019), AI-PET (5 items, Malmström et al. 2023), AI-STB (4

Table 2 Demographic characteristics of PhD Students.

Characteristic	*Items*	*Frequency*	*Percent*
Gender	Female	157	0.43
	Male	204	0.57
Age	23-28	210	0.58
	29-34	105	0.29
	35-41	42	0.12
	42+	4	0.01
PhD Field	Biology	74	0.20
	Physical	68	0.19
	Statistics	57	0.16
	Chemistry	71	0.20
	Mathematics	91	0.25
OpenAI Subscription	ChatGPT Free (Limited access to GPT-4o)	250	0.69
	ChatGPT Plus (GPT 4, GPT 4o, GPT 4o mini)	111	0.31
ChatGPT Use Frequency (per week)	1-2 time	60	0.17
	3-4 time	93	0.26
	More Than 5	208	0.58

items, Y.-Y. Wang and Wang, 2022), PEOU (4 items, Davis, 1989), PU (4 items, Davis, 1989) and BI (3 items, Venkatesh and Davis, 2000). The survey was designed using a 5-point Likert scale and was arranged to be completed within 15–20 min.

Pre-testing. In order to ensure the content validity of the questionnaire, the opinions of four academicians who are experts in the field of AIED were obtained. In line with the expert feedback, conceptual clarity was ensured, linguistic expressions were improved, and the order of the items was rearranged. Following this process, a pilot study was conducted with 15 PhD students in the field of natural sciences in order to assess the comprehensibility, applicability, and contextual validity of the scale. This pre-testing process is in line with the good practices recommended to increase the validity of questionnaire applications, especially in the context of new learning technologies (Wang et al. 2024a). The questionnaire form was finalized in light of the feedback obtained as a result of the pilot application. Detailed survey items and references for each structure are presented in Appendix 1.

Data analysis. This exploratory study focuses on explaining the effects of the dimensions included in the conceptual model and the causal relationships to predict the BI of PhD students regarding the use of ChatGPT. In testing structural relationships, the data did not meet the multivariate normality assumption (Mardia's MVN test, $p < 0.001$) due to the model's complexity and exploratory nature. Therefore, the PLS-SEM method was used to test the proposed conceptual framework in the study, and the analyses were conducted using SmartPLS (v.4.1.0.8). PLS-SEM is preferred to understand theoretical models better when the data do not meet the assumptions of normal distribution, the model includes formative constructs, and there are higher-order constructs. For these reasons, PLS-SEM, a multivariate analysis technique (Hair et al. 2021) was used in the study to examine both the measurement and structural models.

Result

In this study, a two-stage method was used to evaluate the extended TAM; this method includes both measurement and structural model analysis. In the first stage, it was defined as measurement model analysis, and the validity and reliability of the model were examined (Hair et al. 2019). The following stage analyzed the structural model to evaluate the research hypotheses.

Measurement model evaluation. The measurement model has been evaluated based on indicator reliability, internal consistency reliability, convergent validity, and discriminant validity (Hair et al. 2021). Table 3 shows the structures, items, factor loadings, Cronbach's alpha (*CA* α), composite reliability (*CR*), average variance extracted (*AVE*), and Variance Inflation Factor (*VIF*) values from the measurement model.

Convergent validity. In the examination of the validity and reliability of the measurement model, the standardized factor loadings of all items in the model were first examined, and the standardized factor loadings were found to be greater than 0.60 and statistically significant ($p < 0.001$). This result has been interpreted as an indication of high convergent validity. (MacCallum et al. 2001; Hair et al. 2019). For scale reliability, it is expected that the *CA* (α) and *CR* values are above 0.70 (Hair et al. 2019) and the *AVE* value is greater than 0.50. (Fornell and Larcker, 1981; Hair et al. 2019). In the study, it was determined that all latent constructs had *CA* (α) values (the range of 0.782–0.876), *CR* values (the range of 0.791–0.883), and *AVE* values (the range of 0.605–0.730) above the specified threshold values (Table 3), and that the measurement model's reliability, construct reliability, and convergent validity were at an acceptable level. This result has provided evidence regarding the accuracy and consistency of the structures included in the model (Hair et al. 2021).

Additionally, the linearity condition of the measurement model was statistically examined using the *VIF*. Quantitatively measures the degree of collinearity between the indicators in the *VIF* formative measurement model and the set of predictive structures in the structural model, and *VIF* values of 3 and above are considered indicative of multicollinearity (Sarstedt et al. 2021). When Table 3 is examined, it is concluded that there is no multicollinearity among the items, as the *VIF* values for all items range from 1.430 to 2.842.

Discriminant validity. In examining the validity and reliability of the measurement model, the standardized Convergent validity assessment of all items in the model was conducted first, followed by the analysis of the model's discriminant validity. Discriminant validity refers to "the high loading of a scale's items on their constructs and their differentiation from other constructs" (Hair et al. 2019, p.9). The study assessed validity using the Heterotrait-Monotrait (HTMT) ratio and the Fornell-Larcker Criterion (FLC). The HTMT value between two latent constructs being below 0.90 (Hair et al. 2019) statistically indicates that discriminant validity is achieved between these two constructs. The results given in Table 4 show that all HTMT values between the latent structures are below 0.90. FLC, on the other hand, demonstrates discriminant validity by showing that the square root of a dimension's AVE value is greater than its correlations with other dimensions. (Hair et al. 2019, p. 10). According to the FLC results presented in Table 5, the square root of each latent dimension's AVE was greater than the correlations with other dimensions. These findings indicate that the measurement model successfully achieves discriminant validity.

Table 3 Constructs, items, factor loads (FL), VIF, t- value, *CA* (α), *CR*, and *AVE*.

Construct	*Item*	*FL*	*VIF*	*t -value*	*CA* (α)	*CR*	*AVE*
SI	SI1	0.840	2.083	11.242	0.876	0.883	0.730
	SI2	0.812	1.938	8.969			
	SI3	0.862	2.222	13.533			
	SI4	0.901	2.842	17.226			
PEN	PEN1	0.734	1.483	5.578	0.782	0.791	0.605
	PEN2	0.848	1.843	10.091			
	PEN3	0.754	1.430	8.058			
	PEN4	0.770	1.529	8.492			
AI-SE	AI-SE1	0.835	2.084	12.675	0.875	0.879	0.727
	AI-SE2	0.843	2.083	16.165			
	AI-SE3	0.842	2.033	16.603			
	AI-SE4	0.889	2.551	22.832			
AI-PET	AI-PET1	0.758	1.718	9.504	0.849	0.869	0.625
	AI-PET2	0.832	1.955	17.847			
	AI-PET3	0.719	1.545	10.704			
	AI-PET4	0.760	1.705	10.471			
	AI-PET5	0.874	2.610	16.206			
AI-STB	AI-STB1	0.813	1.777	12.381	0.830	0.838	0.662
	AI-STB2	0.865	2.069	18.018			
	AI-STB3	0.780	1.654	10.425			
	AI-STB4	0.795	1.760	11.308			
PEOU	PEOU1	0.810	1.876	25.150	0.874	0.881	0.726
	PEOU2	0.864	2.282	37.706			
	PEOU3	0.840	2.075	28.915			
	PEOU4	0.892	2.583	49.462			
PU	PU1	0.835	1.997	25.386	0.868	0.874	0.717
	PU2	0.803	1.824	22.911			
	PU3	0.852	2.178	33.754			
	PU4	0.894	2.622	46.705			
BI	BI1	0.901	2.021	40.179	0.807	0.830	0.721
	BI2	0.823	1.686	18.454			
	BI3	0.821	1.694	19.010			

Table 4 Discriminant Validity: HTMT ratio.

	AI-PET	AI-SE	AI-STB	BI	PEN	PEOU	PU	SI
AI-PET								
AI-SE	0.163							
AI-STB	0.086	0.088						
BI	0.554	0.521	0.453					
PEN	0.078	0.047	0.092	0.280				
PEOU	0.551	0.612	0.493	0.849	0.403			
PU	0.493	0.581	0.539	0.777	0.098	0.827		
SI	0.079	0.063	0.051	0.103	0.079	0.053	0.437	

Table 5 Discriminant Validity: Fornell-Larcker Criterion.

	AI-PET	AI-SE	AI-STB	BI	PEN	PEOU	PU	SI
AI-PET	**0.791**							
AI-SE	0.142	**0.853**						
AI-STB	0.040	−0.069	**0.814**					
BI	0.469	0.442	−0.377	**0.849**				
PEN	0.018	−0.024	0.010	0.227	**0.778**			
PEOU	0.483	0.537	−0.423	0.726	0.339	**0.852**		
PU	0.429	0.508	−0.462	0.661	0.070	0.724	**0.847**	
SI	−0.068	0.040	−0.032	0.087	0.057	0.031	0.383	**0.854**

Model fit indices. PLS-SEM, unlike classical SEM approaches, recommends the use of standardized root mean square residual (SRMR) and Bentler-Bonett normed fit index (NFI) for evaluating model fit (Henseler et al. 2016). SRMR = 0.057 and NFI = 0.835 were found to evaluate the approximate fit of the estimated model. Hu and Bentler (1999) suggested a threshold value of 0.08 for SRMR as acceptable. In terms of NFI, values above 0.90 generally represent an acceptable fit (Henseler et al. 2016), and a NFI value close to 1.00 is considered a sufficient criterion for evaluating model fit. These results indicate that the

Table 6 Model Fit.

	Saturated model	Estimated model
SRMR	0.049	0.057
d_ULS	1.248	1.697
d_G	0.503	0.562
NFI	0.846	0.835

approximate fit of the measurement model is at an acceptable level (see Table 6).

Structural models' evaluation. In order to ascertain the significance level for each of the theoretical paths presented in the research model, we conducted a bootstrapping procedure with 5000 subsamples based on the robust measurement model. In the evaluation of the structural model, the following methods proposed by Hair et al. (2021) were examined in order: assessment of the model's collinearity status, the magnitude and, significance of path coefficients ($\beta - values$), R^2 values of endogenous variables, Stone-Geisser's Q^2 value, and effect size f^2 methods.

Analysis of collinearity. The collinearity assessment for the structural model, while similar to the collinearity assessment in the formative measurement model, is calculated using the scores of latent variables for predictor dimensions in partial regression to determine the VIF values (Hair et al. 2019). Ideally, *VIF* values should be close to 3 and lower. In cases where multicollinearity is an issue, creating theoretically supported higher-order models is a commonly used method. (Hair et al. 2019, 2022). For the structural model, the VIF values ranged from 1.001 to 2.104, and these results indicate that multicollinearity among the predictor constructs in the structural model does not pose a problem (Table 7).

Path coefficients. The path coefficients included in the model represent the relationships proposed in the research hypotheses. The statistical assessment of path coefficients is evaluated using the bootstrap method, and the values of the coefficients within the range of "−1 to +1" are examined. (Hair et al. 2020). In the structural model, 9 path coefficients were found to be statistically significant at the 99% confidence level, while one (PU → BI) was significant at the 95% confidence level, with path coefficients ranging from −0.412 to 0.520 (Table 7). The significance of path coefficients indicates that the hypothesized relationships are statistically reliable, while their importance shows how strong these relationships are in practice.

R^2 *values*. The next step in evaluating the structural model is to examine the R^2 values of the endogenous variables. To determine the model's explanatory power, the commonly used R^2 measures the explained variance in each of the endogenous structures (Hair et al. 2021). R^2 values of 0.75, 0.50, and 0.25 are considered high, medium, and low levels of explanatory power, respectively (Henseler et al. 2009; Sarstedt et al. 2021). In this study, the R^2 values of the three latent constructs in the structural model were calculated as $R^2_{BI} = 0.565$, $R^2_{PU} = 0.717$, and $R^2_{PEOU} = 0.744$ ($p < 0.001$). The R^2 value regarding PhD students' use of ChatGPT indicates how well behavincluded ioral intention explains the changes in the actual use of ChatGPT. The latent structures in the model explain 56.5% of the variation in PhD students' use of ChatGPT, indicating that the model has a moderate explanatory power. According to the analysis, the R^2 values calculated provide strong evidence that the model is consistent with the research context.

Predictive relevance (Q^2). Stone-Geisser Q^2 value (Stone, 1974; Geisser, 1974) was used as a criterion to evaluate the predictive relevance of the structural model. The Q^2 values being greater than zero is considered an indicator of the predictive relevance of latent endogenous structures (Hair et al. 2019). However, when the Q^2 values are "greater than 0, 0.25, and 0.50, the PLS-path model shows predictive relevance at low, medium, and high levels, respectively." (Hair et al. 2020, p. 107) In the structural model, the endogenous latent constructs were found to have values of $Q^2_{BI} = 0.540$, $Q^2_{PU} = 0.682$, and $Q^2_{PEOU} = 0.736$, indicating that the model has strong predictive capability, and this result was interpreted as the model accurately reflecting reality without error. (Hair et al. 2021).

Effect size (f^2). The effect of removing a specific predictor construct from the structural model on the R^2 value of an endogenous construct is called the f^2 effect size and "is similar to the size of the path coefficients" (Hair et al. 2021, p. 119). Effect size f^2 is classified as small ($0.02 \le f^2 < 0.15$), medium ($0.15 \le f^2 < 0.35$), and large ($f^2 \ge 0.35$) effects (Hair et al. 2020, p. 109). The f^2 values given in Table 7 indicate that some exogenous variables have a more pronounced effect on endogenous variables, while others contribute at a small to moderate level. This evaluation reveals the complex interactions between the latent structures in the model.

Evaluation of hypotheses. The results of the hypothesis tests are given in Table 7 and shown in Fig. 2, and the hypotheses have been discussed by evaluating the statistical significance of the structural parameters in the model. The analysis results have shown that social influence has a statistically significant positive effect $(H1: \beta = 0.375, t = 12.613, p < 0.001)$ and a large effect size (magnitude of the effect) ($f^2 = 0.492$) on PhD students' perceived benefits of using ChatGPT. The perceived entertainment beliefs regarding the use of ChatGPT have a significant effect on the perceived ease of use ($H2: \beta = 0.346, t = 10.947, p < 0.001$), and the effect size ($f^2 = 0.467$) is found to be large. Similarly, AI-SE was found to have a statistically significant effect on PEOU ($H3: \beta = 0.457, t = 13.998, p < 0.001$) and f^2 =0.792 was considered large. It has been observed that the AI ethics perceived by students have a significant positive effect on the PEOU ($H4: \beta = 0.428, t = 13.995, p < 0.001$) and the PU ($H5: \beta = 0.224, t = 4.417, p < 0.001$) of ChatGPT, respectively. The effect size is interpreted as large for $f^2_{AI-PET \to PEOU} = 0.699$ and small for $f^2_{AI-PET \to PU} = 0.122$.

The AI sociotechnical blindness was expected to hypothetically have a negative impact on the perceived ease of use and benefits of using ChatGPT, and this assumption has been confirmed. The effect of AI-STB on PEOU was $\beta = -0.412$ ($t = 13.995, p < 0.001$), while its effect on PU was $\beta = -0.248 (t = 4.417, p < 0.001)$, indicating the presence of a significant negative relationship. As one of the pioneering variables defined in the classic TAM, perceived ease of use has shown a positive significant effect on the perceived usefulness of ChatGPT usage ($H8: \beta = 0.500, t = 8.955, p < 0.001$) and on students' behavioral intentions ($H9: \beta = 0.520, t = 6.073, p < 0.001$), with effect sizes determined as $f^2_{AI-PEOU \to PU} = 0.501$ large, $f^2_{PEOU \to BI} = 0.295$ small.

Finally, it has been observed that the perceived usefulness of using ChatGPT has a statistically significant impact on behavioral intention ($H10: \beta = 0.285, t = 2.145, p < 0.05$), and the effect size has been evaluated as small $(f^2 = 0.089)$.

Discussion

This study has addressed the behavioral intentions and factors determining the use of ChatGPT for educational purposes among

Table 7 Structural estimates (hypothesis test).

Hypothesis	*β*	*stdev*	f^2	*VIF*	*t*-statistics	*p*-values	*Decision*
H1 : SI → PU	0.375	0.034	0.492	1.010	12.613	0.000	Supported
H2 : PEN → PEOU	0.346	0.038	0.467	1.001	10.947	0.000	Supported
H3 : AISE → PEOU	0.457	0.036	0.792	1.027	13.998	0.000	Supported
H4 : AIPET → PEOU	0.428	0.035	0.699	1.453	13.995	0.000	Supported
H5 : AIPET → PU	0.224	0.054	0.122	1.024	4.417	0.000	Supported
H6 : AISTB → PEOU	−0.412	0.035	0.658	1.007	13.784	0.000	Supported
H7 : AISTB → PU	−0.248	0.050	0.161	1.346	5.288	0.000	Supported
H8 : PEOU → PU	0.500	0.061	0.501	1.760	8.955	0.000	Supported
H9 : PEOU → BI	0.520	0.107	0.295	2.104	6.073	0.000	Supported
H10 : PU → BI	0.285	0.111	0.089	2.104	2.145	0.032	Supported

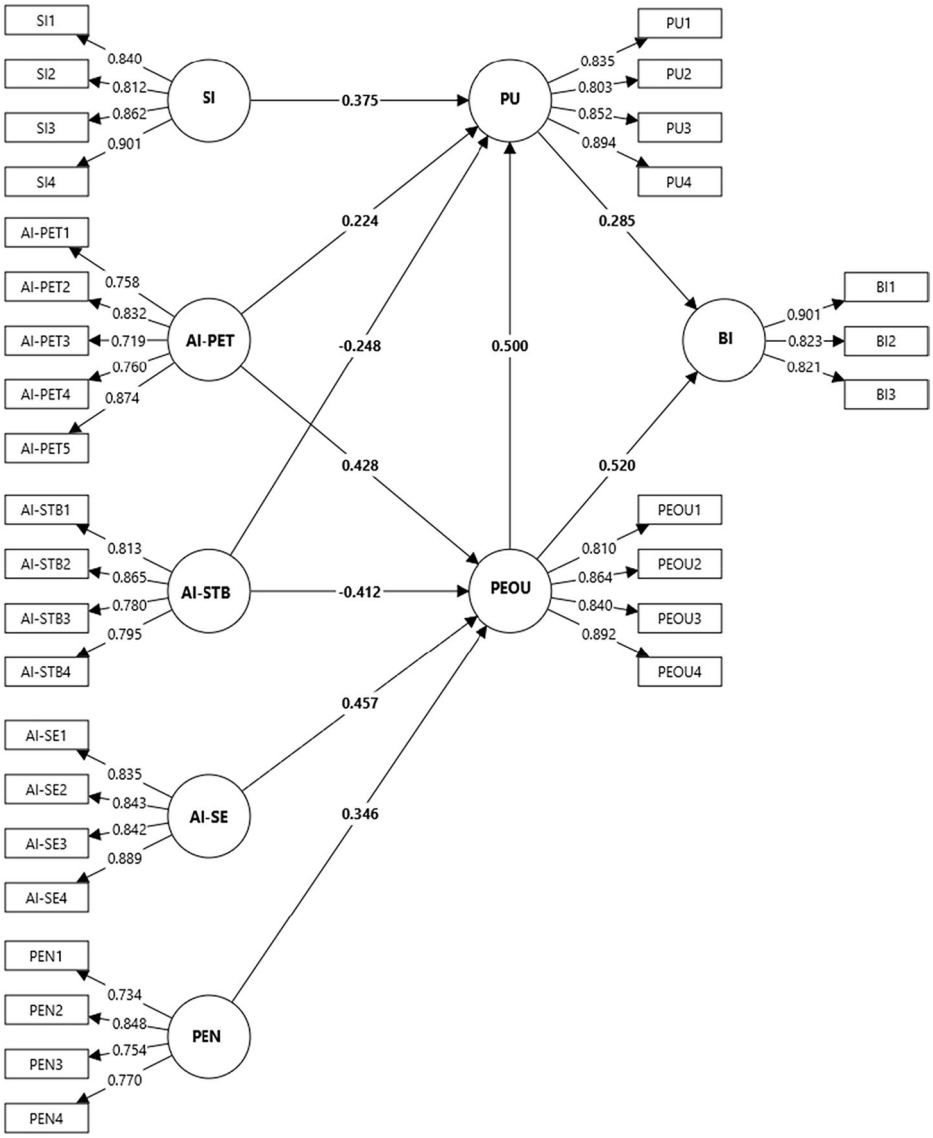

Fig. 2 Structural Model: PLS-SEM algorithm results.

doctoral students in the field of natural sciences at HEIs in Turkey, using the extended TAM and the PLS-SEM approach. The results confirmed that the behavioral intentions of PhD students to use ChatGPT for educational purposes were validated by the conceptually defined research model and that the latent structures of PEOU and PU positively impacted BI. The findings were supported by the literature (Güner et al. 2024; Shao et al. 2024; Almogren et al. 2024; Saihi et al. 2024).

The study has shown that social influence has a significant impact on the PU of using ChatGPT. The significant positive effect of social influence on PU indicates that doctoral students' beliefs about the usefulness of ChatGPT are positively influenced

by their social circles. This finding is consistent with the results reported by Güner et al. (2024), Boubker (2024) and Zhang et al. (2023), which indicates that the use of AI-based technologies like ChatGPT enhances perceived benefits through social influence.

The study has shown that the enjoyment perceived by doctoral students is an important factor in explaining the ease of use of ChatGPT. It has been predicted that doctoral students enjoy using ChatGPT, and this positively enhances their beliefs regarding its ease of use. This result deals with Zhang et al. (2023), Algerafi et al. (2023), Dehghani and Mashhadi (2024) and, Abdalla (2024), which explains the relationship between PEN and PEOU in ChatGPT usage. At the same time, the obtained result aligns with the theory that the perceived entertainment factor plays a motivational role in the technology adoption processes and affects the perceived ease of use (Compeau and Higgins 1995; Venkatesh and Bala 2008). As with perceived enjoyment, it has been determined that AI-SE is also positively related to PEOU. This result shows that PhD students' belief in their ability to use AI technologies strengthens their perception of the ease of use of ChatGPT. At the same time, this result also shows that PhD students have a high belief in their ability to use ChatGPT in education and receive assistance. This finding also corroborates the findings reported by Al Darayseh (2023), Shao et al. (2024), Nja et al. (2023) and Algerafi et al. (2023), which states that AI-SE affects the PEOU of ChatGPT.

The AI-PET belief expresses the doctoral students' views on the moral and ethical implications of using ChatGPT for academic purposes. Research findings have supported that students perceived ethical beliefs, PEOU, and PU regarding the use of ChatGPT have a significant positive impact. In the literature, studies integrating the perceived AI ethics framework into the TAM are limited (Shao et al. 2024). Our study addressing this gap considers the concept of perceived ethics as a premise for how students' perceptions of the ease of use and usefulness structures of educational ChatGPT shape their positive beliefs. Using ChatGPT with an ethics-focused approach will help shape more accurate behavioral intentions. The findings obtained in the study indicate that students have a clear understanding of avoiding ethical dilemmas when using ChatGPT for the purpose of producing educational content, and this finding is consistent with the results of other studies on the ethical use of AI-based technologies like ChatGPT (Acosta-Enriquez, et al. 2024b; Farhi et al. 2023; Kajiwara and Kawabata, 2024; Malmström et al. 2023; Salifu et al. 2024; Usher and Barak, 2024).

AI anxiety is widely discussed in the literature, focusing on the use of AI technologies. In this study, the "sociotechnical blindness" dimension of the AI anxiety scale was developed in four dimensions by Wang and Wang (2022), has been integrated into the conceptual model framework of our study. Johnson and Verdicchio (2017a) state that, unlike other technophobias, AI anxiety can stem from misconceptions about AI technology, confusion about autonomy, and sociotechnical blindness. A type of "sociotechnical blindness" that conceals the fundamental role of humans in the design and implementation stages of an AI system (Johnson and Verdicchio 2017a). Sociotechnical blindness, by paving the way for the emergence of unrealistic scenarios that exclude the human factor (Johnson and Verdicchio 2024), causes the formation of belief patterns in which individuals fail to recognize the social and ethical implications of the development and use of AI technology. For these reasons, it is a priori assumed that doctoral students possess a higher level of ability to interpret AI anxiety, considering their educational levels and career plans. The AI sociotechnical blindness (AI-STB) perceived by doctoral students was expected to have a significantly negative impact on their perceived ease of use and usefulness of using ChatGPT, and this was confirmed by the findings. However, the total indirect effect of AI-STB on behavioral intention was calculated as $\beta_{AI-STB \to BI} = -0.343$ $(t = 14.701, p < 0.001)$. In the literature, users' AI anxiety has been explained through various question forms and numbers; TAM (Al Darayseh, 2023; Algerafi et al. 2023; Chen et al. 2024; Guner and Acarturk, 2020; Saif et al. 2024; Sallam et al. 2024; Wang et al. 2021), UTAUT (Budhathoki et al. 2024; Li 2024) and other models ((Kaya et al. 2024; Wang et al. 2024b; Demir-Kaymak et al. 2024) and it has been reported that it has a negative impact on user behaviors. As a result, the opinions of PhD students regarding the usefulness and ease of ChatGPT are negatively affected by AI-STB and predicted based on findings that students have a significant level of concern.

The study revealed that the PEOU, PU, and BI of doctoral students when evaluating the use of ChatGPT as a learning tool align with the TAM theory. Within the scope of the study, it has been confirmed once again that PEOU is a key determinant of technology acceptance (Davis and Venkatesh,1996). This finding shows that the simple and effortless usability of ChatGPT increases doctoral students' perception of its usefulness for learning purposes and their BI to use it. It has been confirmed in accordance with the literature that PU also has a positive effect on the BI of doctoral students towards the use of ChatGPT. However, within the classical TAM framework, PU is expected to have a stronger impact on BI than PEOU (Venkatesh and Davis, 2000). Algerafi et al. (2023) found that the effects of PEOU and PU on BI were equal ($\beta = 0.203$); Pan et al. (2024) investigated the key factors affecting university students' willingness to use AI Coding Assistant Tools (AICATs) and found $\beta_{PEOU \to BI} = 0.468$ ve $\beta_{PU \to BI} = 0.312$; Almogren et al. (2024) obtained similar results to our study $\beta_{PEOU \to BI} = 0.360$ ve $\beta_{PU \to BI} = 0.130$. This situation is explained by the fact that PEOU is a strong predictor of PU (Kampa 2023) and BI in adopting new technologies (Zhou et al. 2022; Lai et al. 2023). One possible reason for this difference is thought to be the specificity of the conditions under which the study was conducted. Our study covers a period when ChatGPT was newly used in HIEs in Türkiye. It seems that intuitive usability may be more decisive than perceived long-term benefits at this early stage of use. The findings reveal that ease of use plays a stronger role in behavioral intention among users who encounter a new technology, such as ChatGPT, for the first time, even among students studying in disciplines closely related to technology, such as natural science. In this context, the study makes a theoretical contribution to the TAM literature by revealing the time-dependent factors that affect the PEOU-PU-BI relationship and user tendencies at the time of initial contact with the technology. In addition to these, the findings obtained in the study are similar to those reported by Algerafi et al. (2023), Zhang et al. (2023), Lai et al. (2023), Güner et al. (2024), Pan et al. (2024), and Zou and Huang (2023) which shows the PU affected by PEOU, and accordingly, these two factors have significant effects on BI in the context of AI chatbots and ChatGPT.

To the end, it has been determined that doctoral students in the field of science possess sufficient basic knowledge about AI, use ChatGPT for learning purposes, and have positive behavioral intentions.

Conclusions

In this study, the extended TAM was used to investigate the behavioral intentions of PhD students in the field of science toward the use of ChatGPT and the factors influencing it. The study used data obtained from 361 students pursuing PhD education in mathematics, physics, chemistry, biology, and statistics at 35 universities in Turkey and applied the PLS-SEM method for data analysis. The results have shown that external structures that positively and negatively affect behavioral intentions towards ChatGPT interact at a causal level. The study also found that the

total indirect effects of social influence, perceived AI ethics, self-efficacy, and perceived entertainment structures on behavioral intentions were significant and positive, while "sociotechnical blindness," which represents perceived AI anxiety, had a significant and negative impact.

The PEOU and PU have mediated the effect of the latent variables that are included in the proposed model to extend TAM on BI, and it has been emphasized that these mediating structures have created significant and positive effects on students' behavioral intentions. The strong behavioral intention of PhD students in the field of natural sciences to use ChatGPT for learning purposes indicates that ChatGPT could enhance its function as an educational tool in PhD education in higher education. Additionally, while all causal relationships indicate that PhD students have a strong intention to use ChatGPT, measures should be taken to mitigate the negative effects of ChatGPT usage on PhD students.

Overall, this study is of great importance in providing a projection of the decisions made by doctoral students in the basic fields of science in Turkey regarding the use of ChatGPT, as well as filling the gap in the literature. The high level of AI ethics possessed by doctoral students, the significant AI anxiety they shape, their levels of AI self-efficacy, and their positive behavioral intentions towards the use of ChatGPT for learning purposes have emphasized the increasing importance of AI in higher education.

The Turkish Higher Education Council (YÖK, 2024b) has taken an important step in this regard by publishing an ethical guide in May 2024, which outlines fundamental values and principles to govern the integration of Generative AI into educational and scientific processes. However, based on research conducted using Turkish university websites (.edu.tr domain), few institutions appear to have implemented or published detailed guidance on AI usage or related ethical principles. This indicates that AI integration into scientific research and publication processes is still in its formative phase.

Both globally and within the Turkish higher education context, there is an urgent need to revise institutional policies and evaluation mechanisms to ensure academic integrity while using AI tools. As emphasized by Lo (2023) and Foroughi et al. (2023) a high level of sensitivity is required in adopting ChatGPT in education, and ethical risks must be actively addressed.

Theoretical implications. The findings underscore the theoretical relevance of extending the Technology Acceptance Model by integrating sociotechnical, affective, and ethical components such as perceived AI ethics, AI self-efficacy, perceived enjoyment, and AI anxiety (conceptualized as sociotechnical blindness). This model extension offers a richer framework to understand doctoral students' behavioral intentions towards using generative AI tools like ChatGPT in academic contexts.

The role of mediating structures—Perceived Usefulness (PU) and Perceived Ease of Use (PEOU)—is particularly notable, as they serve as conduits through which external variables exert their influence on behavioral intention. This supports and enriches earlier findings in TAM-based literature by confirming the central role of PU and PEOU in facilitating technology adoption among advanced learners. In addition, this result suggests that users primarily focus on intuitive usability during the adoption process of new technologies, such as GenAI, and that their perception of functional benefits may develop over time. In addition, the strong negative effects of AI anxiety (AI-STB) on PU and PEOU indicate that not only cognitive but also psychosocial barriers are determinants of technology use. The results obtained regarding the effects of AI anxiety and other external factors included in the extended model reveal the necessity of considering the emotional and perceptual readiness levels of users, especially when using GenAI technologies; in this respect, it expands the theoretical scope of technology acceptance models.

Practical implications. This study has several practical implications for technology developers, educational policymakers, and faculty members. The results highlight the need for proactive strategies that promote the responsible and ethical integration of GenAI tools into doctoral education. In particular, institutions should focus on:

- Raising awareness about ethical AI usage, especially considering the ethical concerns and anxiety identified among doctoral students (Foroughi et al. 2023; Lo, 2023),
- Developing targeted training programs for both students and educators to ensure informed, responsible, and pedagogically meaningful AI use (Lund and Wang, 2023; Farrokhnia et al. 2024),
- Embedding AI literacy into higher education as a component of graduation skills, aiming to improve learning outcomes and preserve academic integrity.

Additionally, the study findings suggest that emotional and cognitive barriers to AI should be considered not only at the level of individual awareness but also in institutional educational policies and instructional design processes. While ethical uncertainties and perceptual concerns arising from AI are also observed, especially among doctoral students, universities need to adopt holistic educational strategies that develop not only technical competence but also ethical thinking, technological critique, and psychological resilience.

Data availability

The datasets analyzed during the current study are available from the corresponding author upon reasonable request.

Received: 17 December 2024; Accepted: 28 July 2025;
Published online: 13 August 2025

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Author contributions

Conceptualization by FU, EK, and HEÇ, methodology by EK, FU, and HEÇ, validation by HEÇ and FU, formal analysis by HEÇ and FU, references by EK and FU, data collection by HEÇ and FU, writing—original draft by EK and FU, review and editing by FU, EK, and HEÇ. All authors have read and agreed on the publication version of the manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

This study was conducted according to the guidelines of the Declaration of Helsinki and approved by the Scientific Research Ethical Committee, Van Yüzüncü Yıl University (reference number: 2023/30-09; date of approval: 26.12.2023). The Academic Committee reviewed and approved the study protocol, including research design and methods, participant eligibility criteria, data collection and confidentiality protection, and informed consent. All online surveys were obtained by taking into account the informed consent of the participants, and all participants completed the survey voluntarily and anonymously.

Informed consent

The study was conducted between May 01 and June 25, 2024. All participants gave informed consent before completing the official survey. At the beginning of each web-based survey, they were informed about the study's objectives, procedures, and confidentiality of their responses, and they were told that the data collected would be used only for academic research. It was also clearly stated that any personal information would be provided anonymously and that they reserved the right to withdraw from the study at any time. All participants clicked the **"Accept"** button to complete the informed consent process, agreeing to participate, have their data used for research purposes, and have anonymous findings published.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1057/s41599-025-05641-x.

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